Osptek Display

TFT LCD SPECIFICATION

Model No:

YDP160B001-V1

Revision Record

Rev No.	Date	Description
A/0	2023-02-03	First Release

osptek®

CONTENTS 目录

Revision Record

Contents

ITEM	PAGE
1. General Specifications	4
2. Pin Assignments	5
3. Electrical Specification	6-7
3.1 Absolute Maximum Ratings	6
3.2 Typical Operation Conditions	7
3.3 Backlight Circuit Characteristics	7
3.4 LCD Current Consumption	7
4. Optical Specification	8
4.1 LCM Optical Characteristics	8
4.2 Measurement system	9
5. Mechanical Drawing	10
6. Reliability Test Items	11
7. Packing and Storage Specification (Reference only)	12 -13
8. Announcements	14

osptek®

1.General Specifications

No.	Item	Specification	Unit	Remark
1	LCD Size	1.6''	inch	-
2	Panel Type	IPS	-	-
3	Resolution	400x RGB x 400	Pixel	-
4	Display Mode	Normally Black		-
5	Number of Colors	16M	-	-
6	Viewing Direction	All	-	Note1
7	NTSC	65%	-	Typ.
8	Contrast Ratio	1200	-	
9	Luminance	600	cd/m2	Typ.
10	Module Size	42.14x44.24(V)x1.96(D)	mm	Note1
11	Panel Active Area	Φ39.84	mm	Note1
12	Pixel Pitch		mm	-
13	Pixel Arrangement	RGB-stripe		-
14	Weight	TBD	g	-
15	Driver IC	ST7797	-	-
16	Driver IC RAM Size		bit	-
17	Light Source	3 White LEDs	-	-
18	Interface	MIPI	-	-
19	Operating Temperature	-20~+70	℃	-
20	Storage Temperature	-30~+85	℃	-

Note 1: Please refer to the mechanical drawing；

osptek®

2.Pin Assignments

No.	Symbol	Function
1	GND	Ground
2	LCD_RST	Chip reset signal.
3	TDN0	MIPI DSI data lane0-
4	NC	No connection.
5	TDP0	MIPI DSI data lane0+
6	NC	No connection.
7	GND	Ground
8	TE	Tearing Effect pin.
9	TCN	MIPI DSI clock lane-
10	GND	Ground
11	TCP	MIPI DSI clock lane+
12	GND	Ground
13	GND	Ground
14	GND	Ground
15	VIO18	Power supply for LCM.
16	VCC28	Power supply for LCM.
17	GND	Ground
18	GND	Ground
19	NC	No connection.
20	NC	No connection.
21	NC	No connection.
22	NC	No connection.
23	LEDK	Power supply-for backlight cathode.
24	LEDA	Back light power supply positive.

Osptek®

3. Electrical Specification

3.1 Absolute Maximum Ratings 极限参数

Item 项目	Symbol	Value	Unit	Remark
Analog Power Supply Voltage 模拟供电电压	VCI	-0.3~+3.3	V	-
Digital Power Supply Voltage 数字电源电压	VDD	-0.3~+3.3	V	-
I/O Power Supply Voltage I/O端口供电电压	IOVCC	-0.3~3.3	V	-

3.2 Typical Operation Conditions 典型工作条件

Item 项目	Symbol	Min.最小	Typ.典型	Max.最大	Unit
Analog Supply Voltage 模拟供电电压	VCI	2.7	2.8	3.3	V
Digital Supply Voltage 数字电源电压	VDD	2.7	2.8	3.3	V
I/O Supply Voltage 接口电压	IOVCC	1.65	1.8/2.8	3.3	V
Input High Voltage 输入高电平	V_{IH}	0.8*IOVCC	-	IOVCC	V
Input Low Voltage 输入低电平	V_{IL}	0	-	0.2*IOVCC	V
Output High Voltage 输出高电平	V_{OH}	0.8*IOVCC	-	-	V
Output Low Voltage 输出低电平	V_{OL}	-	-	0.2*IOVCC	V

3.3 Backlight Circuit Characteristics 背光功耗

Item	Symbol	Min.	Typ.	Max.	Unit
LED Current 背光电流	I_B	-	20	-	mA
LED Voltage 背光电压	Vf	8.4	9	9.6	V
Power Consumption 功耗	P_{BL}	-		-	mW

3.4 LCD Current Consumption 液晶面板功耗

Item	Symbol	Typ.	Max.	Unit
Full Mode 正常模式	VCI	-	-	mA
测试条件：VCI=2.8V，IOVCC=2.8V； Interface 驱动类型：行翻转或者列翻转； TN Type=>All Black Pattern. TN型液晶面板=>黑色画面； IPS Type=>All White Pattern. IPS型液晶面板=>白色画面； Temperature：25℃；温度：室温25摄氏度；				
Sleep Mode 休眠模式	VCI	-	-	uA
测试条件：VCI=2.8V，IOVCC=2.8V； DC/DC converter is enabled. Internal oscillator is started and panel scanning is started. 除IC内部晶振和面板扫描外，其他功能都暂停工作； Temperature：25℃；温度：室温25摄氏度；				

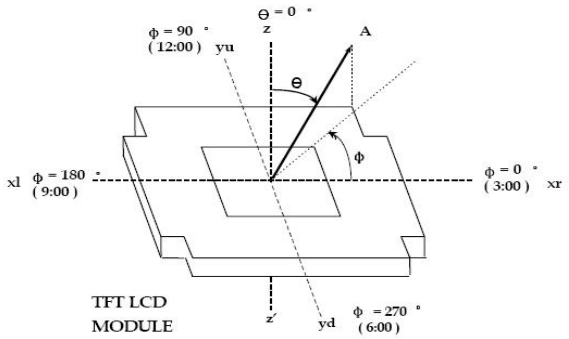
4.Optical Specification

4.1 LCM Optical Characteristics 液晶模组光学特性

Item		Symbol	Condition	Min.	Typ.	Max.	Unit
Viewing Angle Range 视角	Left	θ_L	CR≧10	-	85	-	degree
	Right	θ_R		-	85	-	
	Top	θ_T		-	85	-	
	Bottom	θ_B		-	85	-	
Response Time 响应时间		Ton+Toff	θ=Φ=0°	-	25	35	ms
Contrast Ratio 对比度		CR	θ=Φ=0°	-	1200	-	-
Luminance 亮度		L	θ=Φ=0°	-	600	-	cd/m^2
Color Chromaticity (CIE1931) 色坐标	White	W_x	Normal θ=Φ=0°	-0.02	0.294	+0.02	(1)(4) CF glass C-light
		W_y			0.335		
	Red	R_x			0.660		
		R_y			0.323		
	Green	G_x			0.280		
		G_y			0.571		
	Blue	B_x			0.134		
		B_y			0.132		
Uniformity 均匀度		U_L	θ=Φ=0°	80	-	-	%
Flicker 闪烁		-	-	No Visible			-

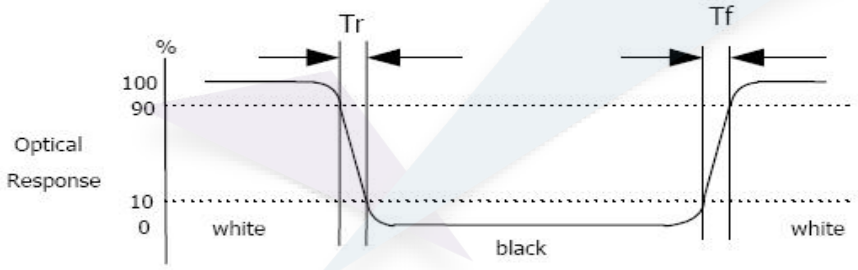
4.2 Measurement system 测量系统

4.2.1 LCM Viewing Angle



Viewing angle is the angle at which the contrast ratio is greater than 10.The angles are determined for the horizontal or x axis and the vertical or y axis with respect to the z axis which is normal to the LCD surface.

4.2.2 Response time



Response time is the time required for the display to transition from white to black (Rising time, Tr) and from black to white (Falling time, Tf) for additional information.

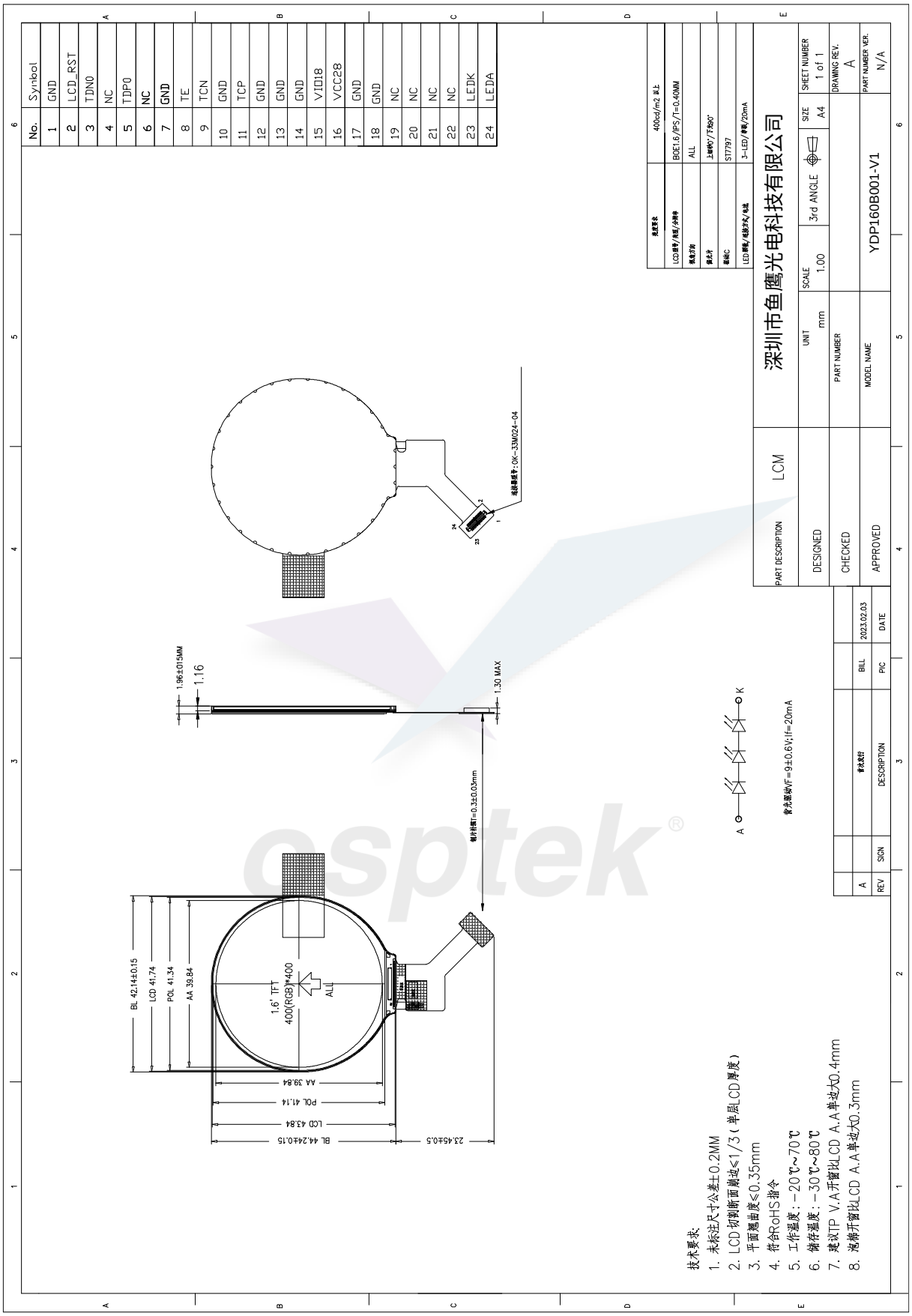
4.2.3 Contrast Ratio (CR)

Contrast Ratio (CR) is defined mathematically as:

$$\text{Contrast Ratio} = \frac{\text{Surface Luminance with all white pixels}}{\text{Surface Luminance with all black pixels}}$$

Surface luminance is the center point across the LCD surface 500mm from the surface with all pixels displaying white.

5. Mechanical Drawing



osptek®

6. Reliability Test Items

Test Item	Test Condition	Test result determinant gist
High temperature storage	80℃，240H；	Inspection after 2~4hours storage at room temperature, the sample shall be free from defects: 1.Air bubble in the LCD; 2.Non-display; 3.Glass crack; 4.The electrical characteristics requirements shall be satisfied.
Low temperature storage	-30℃，240H；	
High temperature operation	70℃，240H；	
Low temperature operation	-20℃，240H；	
High temperature /humidity	60℃±3℃,90%±3%RH，240H；	
Thermal Shock	-30℃/0.5h~+80℃/0.5h for a total 24 cycles；	
Vibration Test	Frequency 10Hz~55Hz~10Hz Amplitude：1.5mm, X，Y，Z direction for total 1H；(Packing condition)	
ESD test	±2KV, Human Body Mode, 150pF/330Ω； ±8KV, Air Mode, 150pF/330Ω；	

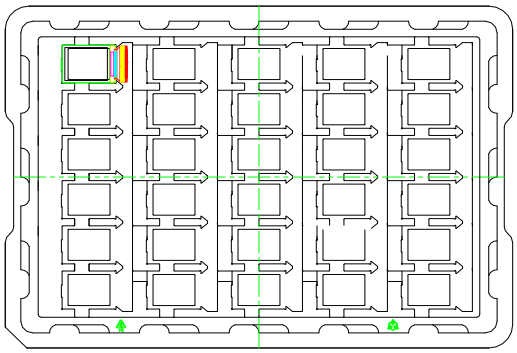
Remark:

1. The test samples should be applied to only one test item.
2. Sample size for each test item is 2pcs.
3. Failure Judgment Criterion: Basic Specification, Electrical Characteristic, Mechanical Characteristic, Optical Characteristic.

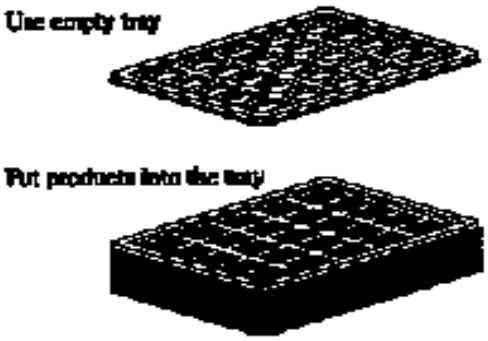
7. Packing and Storage Specification(Reference Only)

7.1 Packing Method

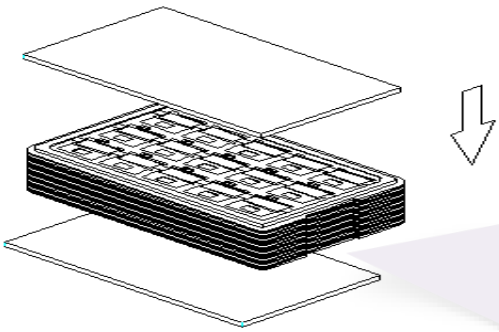
(1)



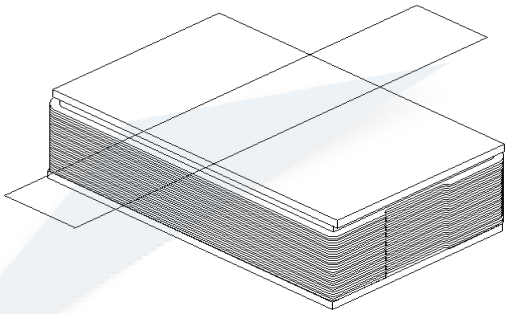
(2)



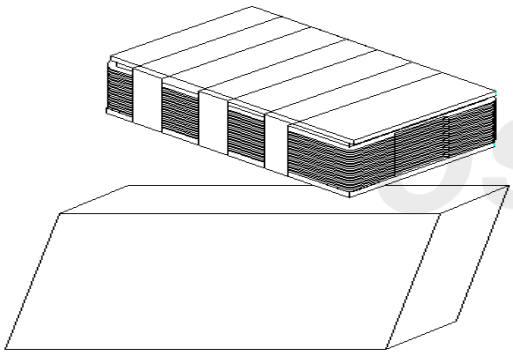
(3)



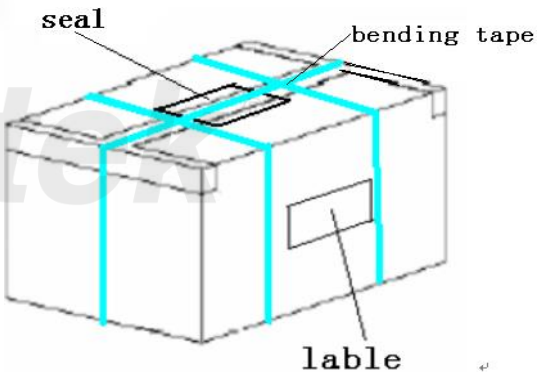
(4)



(5)



(6)



1. Put module into tray cavity. .
2. Tray stacking. .
3. Put 1 foam under the tray stack and 1 foam above. .
4. Fix the cardboard to the tray stack with adhesive tape. .
5. Put the tray stack into carton. .
6. Carton sealing with adhesive tape. .

7.2 Storage Method

1. Store in an ambient temperature of 23°C±5°C, and in a relative humidity of 55%±15%. Don't exceed 12 months and expose to sunlight or fluorescent light.
2. Store in a clean environment, free from dust, active gas, and solvent.
3. Store in antistatic container.

osptek®

8. Announcements

1. Do not attempt to disassemble or process the LCD module.
2. Do not make extra holes on the printed circuit board, modify its shape or change the positions of components to be attached。
3. Except for soldering the interface, do not make any alterations or modifications with a soldering iron; Ensure welding temperature at 320 ° C to 350 ° C, the welding time control within the 10 s, welding note don't stay too long in the same place to avoid scald FPC.
4. Other matters in not clear before use, please contact our staff to guide.
5. Under the specified electrical parameters, the ambient temperature is 23 ° C ± 5 ° C, the humidity is 55%± 15%, and the hardware is not damaged. The normal service life is 30,000-50,000 hours.

osptek®